

Session 6: CPSA Goals

Slides prepared by UMBC Protocol Analysis Lab

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Common Security Goals

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Common Security Goals

- ▶ **Confidentiality:** In a two-party communication protocol, confidentiality holds if no third party is able to access the information sent.
- ▶ **Agreement:** In a two-party communication protocol, agreement holds if both parties agree on some set of data values in the protocol. These values are then used to derive the key.

Other common security goals

1. Aliveness
2. Weak Agreement
3. Non-injective Agreement
4. A specific relay party cannot read the message
5. The message must be timestamped
6. Perfect forward secrecy
7. Crypto primitives are resistant to a predefined attack

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Confidentiality and Agreement (on the secret) needed.

CPSA Goals

- ▶ Protocol goals as mathematical statements
- ▶ Find strands that satisfy (or don't satisfy) some set of properties
- ▶ May reduce time taken to terminate

Some more tips for reading specifications

- ▶ FIND THE SECURITY GOALS
 - ▶ Are additional SGs needed?
- ▶ Modularize the spec into blocks of messages
 - ▶ “In the first set of exchanges”
 - ▶ “[party1] preshares [data] with [party2]”
- ▶ Ignore nonsense
 - ▶ “[protocol] does not consider [niche usecase] because. . .”
 - ▶ “We assume clients have access to an ALU” (structural vs. implementation)
 - ▶ “Our conformance program handles...”

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- ▶ Let's try modeling it!

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Survey

Fill in details about the survey here!